

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

☎ (587) 879-4626 | ✉ tkuipers123@gmail.com | 🏠 tkuipers.ca | 📱 [tkuipers](#) | 🌐 [tyler-kuipers](#)

About Me

I've spent 10 years building backend systems, with the last 3 focused on cloud-native architectures and platform engineering. I've built products from scratch (including an award-winning insurance product that generates \$4.3M annually), architected multi-account AWS systems, and created developer platforms that made teams 60x faster. I work primarily in Python and TypeScript on AWS, building microservices, APIs, and infrastructure automation. I ship fast, iterate based on user feedback, and build tools that multiply team effectiveness. I'm excited about using modern development tools and AI assistants to accelerate delivery while maintaining quality and security.

Work Experience

Alberta Motor Association

Edmonton, Alberta

SOFTWARE ARCHITECT

Jan 2023 - Present

- Architected modernization of document generation system from legacy C# to containerized microservices on EKS. Integrated Datadog for metrics, logging, and alerting. Led 2-3 engineers. Reduced environment provisioning from weeks to minutes. (Kubernetes, Docker, Datadog, Helm, CI/CD)
- Architected technical infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts with external partner integration. 10 Lambda functions, multi-million dollar annual revenue, 2.8M events processed, 99.9% uptime. Won 2024 Guidewire Innovation Award. (AWS, Terraform, Lambda, Python)
- Built self-service infrastructure platform that reduced environment provisioning from 2-3 weeks to 2 hours (60x improvement). Includes database snapshotting for production issue diagnosis. Built solo, adopted by 15 engineers across 3 teams. (Python, Terraform, AWS, Jenkins, Docker)
- Own API architecture for platform serving 5 internal teams and external partners. Approve endpoint additions, maintain design standards, and ensure consistency across services. (Python, REST, OpenAPI)
- Set coding standards and maintain technical documentation for 12-person team. Author RFCs for cross-team initiatives. Member of Architecture Steering Committee reviewing proposals org-wide.
- Mentor engineers on debugging complex distributed systems issues. Trained 5-6 developers on TDD methodologies.

Arterys Inc. (Acquired by Tempus Labs)

Calgary, Alberta (Remote)

SOFTWARE DEVELOPER

Aug 2021 - Aug 2022

- Built backend services running medical AI algorithms, optimizing for low latency and high throughput. (TypeScript, Node.js, AWS)
- Improved algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, identifying and resolving issues before customer impact.

Alberta Motor Association

Edmonton, Alberta

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Built automated test framework that reduced defects by 30% on subsequent releases. Framework adopted org-wide, still in use 4+ years later. (Java/Gosu, JUnit)
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local dev environments.
- Led 5-6 person development team using TDD methodology. Delivered projects on time with no critical defects post-launch.

Pleasant Solutions

Edmonton, Alberta

SOFTWARE DEVELOPER

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

Halifax, Nova Scotia

SOFTWARE ENGINEER ASSOCIATE

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

The King's University

Edmonton, Alberta

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

May 2014 - Sep 2016

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.

Education

Skills

Rapid Product Delivery	Built greenfield products from initial design to production in 6 months. Created platforms that made teams 60x faster. Focus on shipping fast, iterating based on feedback, and enabling rapid experimentation.
Cloud-Native Architecture	Design scalable, secure AWS architectures (Lambda, EKS, RDS, S3, IAM). Built systems serving 5 teams with 99.9% uptime and \$4.3M in annual transactions. Experience with microservices, APIs, and containerization (Docker, Kubernetes).
Cross-Functional Collaboration	Coordinate across engineering, product, and business stakeholders. Translate technical concepts for non-technical audiences. Drive alignment on multi-team initiatives with internal and external partners.
Engineering Velocity	Built test frameworks (30% defect reduction), infrastructure platforms (60x faster provisioning), and developer tooling. Champion best practices in testing, security, and scalability. Strong advocate for modern development tools and AI-powered coding assistants.

Technologies

Languages	Python, TypeScript, Java, Bash
Backend	REST APIs, Microservices, OpenAPI, CI/CD, Event-Driven Architecture
Cloud & DevOps	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, Route53), Terraform, Kubernetes, Docker, Helm
Observability	Datadog, Elasticsearch, Kibana
Databases	PostgreSQL, Aurora/RDS, Redis
AI/ML Exposure	Integrated ML models into production services (medical imaging AI at Arterys), experience with AI-powered development tools